

Computer Science & IT

Database Management System



Relational Model & Normal Forms

Lecture No. 10



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Recap of Previous Lecture



- ✓ **Topic** Problems because of redundancy in a relation
- ✓ **Topic** Schema refinement (Normalization)
- ✓ **Topic** Properties of decomposition
- ✓ **Topic** Dependency preserving decomposition

Topics to be Covered



- ✓ **Topic** Lossless join decomposition (Natural Join)
- ✓ **Topic** Projection, Selection, and Cross product
- ✓ **Topic** Natural join
- ✓ **Topic** Lossless Natural Join





Topic : Properties of decomposition

While decomposing a relation into sub-relations following two properties must be ensured.

① Decomposition must be dependency preserving decomposition
i.e., All functional dependencies present in the original relation must also be preserved even after decomposing the relation into subrelations.

i.e., after decomposition there must not be any loss of information

② Decomposition must be lossless Join decomposition.
i.e., if we perform the natural join ($\bowtie$) operation on all subrelation of original relation, then we must get the exact same set of tuples as of original relation.

(Natural Join)

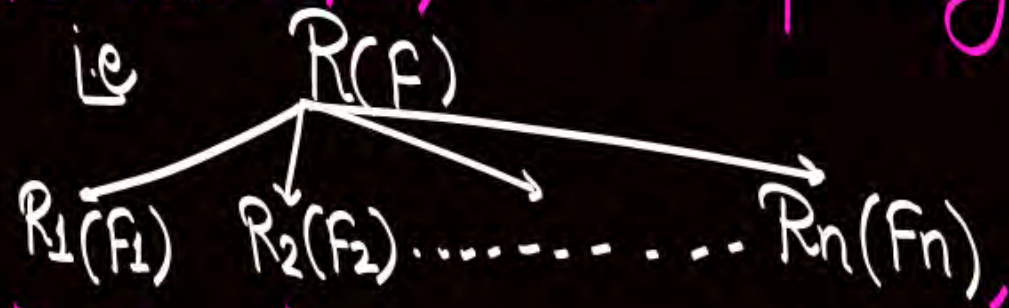
neither in terms of dependency

nor in terms of data

Properties of decomposition

① Dependency Preserving decomposition

Let relation R with FD set F is decomposed into subrelations $R_1, R_2, \dots, R_n$ with FD sets $F_1, F_2, \dots, F_n$ respectively,



then in general,

$$F_1 \cup F_2 \cup F_3 \cup \dots \cup F_n \subseteq F$$

- (i) if, $F_1 \cup F_2 \cup F_3 \cup \dots \cup F_n = F$, then dependency preserving
- (ii) if, $F_1 \cup F_2 \cup F_3 \cup \dots \cup F_n \subset F$, then not dependency preserving
- (iii) if, $F_1 \cup F_2 \cup F_3 \cup \dots \cup F_n \supset F$, not possible

② Lossless join decomposition ✓

Let relation R with FD set F is decomposed into subrelations $R_1, R_2, \dots, R_n$ with FD sets $F_1, F_2, \dots, F_n$ respectively,



In general,

$$R_1 \bowtie R_2 \bowtie \dots \bowtie R_n \supseteq R$$

if, $= R$, then lossless Join

if, $\supset R$, then Lossy Join

if $\subset R$, never possible



Topic : Dependency preserving decomposition

Q. Let $R(A, B, C, D)$ is a relation with FD set $F = \{A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow A\}$ is decomposed into three subrelations $R_1(A, B)$, $R_2(B, C)$, & $R_3(C, D)$.
Check whether the decomposition is dependency preserving or not?

$R(A, B, C, D)$

$F = \{A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow A\}$

$R_1(A, B)$

$R_2(B, C)$

$R_3(C, D)$

$(A)^+ = \{A, B, C, D\}$
 $(B)^+ = \{A, B, C, D\}$
 $F_1 = \{A \rightarrow B, B \rightarrow A\}$

$(B)^+ = \{B, C, D, A\}$
 $(C)^+ = \{C, D, A, B\}$
 $F_2 = \{B \rightarrow C, C \rightarrow B\}$

$(C)^+ = \{C, D, A, B\}$
 $(D)^+ = \{D, A, B, C\}$
 $F_3 = \{C \rightarrow D, D \rightarrow C\}$

$F_1 \cup F_2 \cup F_3 = \{A \rightarrow B, B \rightarrow C, C \rightarrow D, B \rightarrow A, C \rightarrow B, D \rightarrow C\}$

Check whether $F_1 \cup F_2 \cup F_3 = F$ or not?

Case ① Check if F covers $F_1 \cup F_2 \cup F_3$, i.e., check if $F_1 \cup F_2 \cup F_3 \subseteq F$

Case ② Check if $F_1 \cup F_2 \cup F_3$ covers F , i.e., check if $F \subseteq F_1 \cup F_2 \cup F_3$

We don't need to check for this case, It will always hold true

- FDs of F
- ① $A \rightarrow B$, it is directly available in $F_1 \cup F_2 \cup F_3$
 - ② $B \rightarrow C$ it is directly available in $F_1 \cup F_2 \cup F_3$
 - ③ $C \rightarrow D$ it is directly available in $F_1 \cup F_2 \cup F_3$
 - ④ $D \rightarrow A$, $(D)^+ \text{ w.r.t. } F_1 \cup F_2 \cup F_3 = \{D, C, B, A\}$

$\therefore F \subseteq F_1 \cup F_2 \cup F_3$

$A \in (D)^+ \text{ w.r.t. } F_1 \cup F_2 \cup F_3$
 $\therefore D \rightarrow A$ is a member of $F_1 \cup F_2 \cup F_3$

$F_1 \cup F_2 \cup F_3 \subseteq F$
 $F \subseteq F_1 \cup F_2 \cup F_3$
 $\therefore F_1 \cup F_2 \cup F_3 = F$
 $\therefore$ Dep. preserving decomposition

① H.W.:-

Let $R(A, B, C, D)$ is a relation with FD set,

$$F = \{ \underline{AB} \rightarrow \underline{CD}, \underline{D} \rightarrow A \},$$

If relation R is decomposed into two subrelations

$R_1(AD)$ & $R_2(BCD)$, then Check whether the decomposition
is dependency preserving or not.

$R(ABCD)$

$F = \{ AB \rightarrow CD, D \rightarrow A \}$

$R_1(AD)$

$R_2(BCD)$

$(A)^+ = \{A\}$

$(D)^+ = \{D, A\}$

$F_1 = \{D \rightarrow A\}$

$F_2 = \{BD \rightarrow C\}$

$(B)^+ = \{B\}$

$(C)^+ = \{C\}$

$(D)^+ = \{D, A\}$

$(BC)^+ = \{BC\}$

$(BD)^+ = \{BD, AC\}$

$(CD)^+ = \{C, D, A\}$

$F_1 \cup F_2 = \{D \rightarrow A, BD \rightarrow C\}$

① We know $F_1 \cup F_2 \subseteq F$ — eqⁿ ①

② Check if $F \subseteq F_1 \cup F_2$

$D \rightarrow A$ ✓ {Directly present in $F_1 \cup F_2$ }

$AB \rightarrow CD$, $(AB)^+$ w.r.t. $F_1 \cup F_2 = \{A, B\}$

$C, D \in (AB)^+$ w.r.t. $F_1 \cup F_2$ $\therefore AB \rightarrow CD$ is lost. Hence not dep. preserving

② H.W.

Let $R(A, B, C, D, E, F)$ is a relation with FD set
 $F = \{ \underline{AB} \rightarrow C, \underline{BC} \rightarrow A, \underline{AC} \rightarrow B, B \rightarrow D, AD \rightarrow E, E \rightarrow F \}$

Which of the following decomposition is dependency Preserving decomposition

(i) $\mathcal{D}_1 = \{ R_1(A, B, C), R_2(A, C, D), R_3(A, C, E, F) \}$

$$F_1 = \left\{ \begin{array}{l} AB \rightarrow C \\ BC \rightarrow A \\ AC \rightarrow B \end{array} \right\}$$

$$F_2 = \{ AC \rightarrow D \}$$

$$F_3 = \left\{ \begin{array}{l} AC \rightarrow EF \\ E \rightarrow F \end{array} \right\}$$

FDs of set F that are lost,
 $B \rightarrow D, AD \rightarrow E$

$$(AD)^+ = \{ \cancel{A}, \cancel{D}, \cancel{E}, \cancel{F} \}$$

② H.W.

Let $R(A, B, C, D, E, F)$ is a relation with FD set
 $F = \{ AB \rightarrow C, BC \rightarrow A, AC \rightarrow B, B \rightarrow D, AD \rightarrow E, E \rightarrow F \}$

Which of the following decomposition is dependency Preserving decomposition

(ii) $\mathcal{D}_2 = \{ R_1(A, B, C) \quad R_2(E, F), R_3(\tilde{A}, \tilde{B}, \tilde{D}, \tilde{E}) \}$

$$F_1 = \left\{ \begin{array}{l} AB \rightarrow C \\ AC \rightarrow B \\ BC \rightarrow A \end{array} \right\}$$

$$F_2 = \{ E \rightarrow F \}$$

$$F_3 = \left\{ \begin{array}{l} B \rightarrow D \\ AD \rightarrow E \end{array} \right\}$$

$$F_1 \cup F_2 \cup F_3 = F$$

Hence dependency Preserving decomposition



Topic : Lossless Join decomposition

- If we decompose a relation R with FD set F into sub-relations $R_1, R_2, \dots, R_n$ with FD sets $F_1, F_2, \dots, F_n$ respectively, then for this decomposition to be called lossless join decomposition following property must hold true.

➤

$$R_1 \bowtie R_2 \bowtie \dots \bowtie R_n = R$$

if $\supset R$, then lossy.



Topic : Lossless Join decomposition

Let relation R is decomposed into sub-relations $R_1, R_2, \dots, R_n$

In general, $R_1 \bowtie R_2 \bowtie \dots \bowtie R_n \supseteq R$

if, $R_1 \bowtie R_2 \bowtie \dots \bowtie R_n = R$ then, Lossless join decomposition

if, $R_1 \bowtie R_2 \bowtie \dots \bowtie R_n \supset R$ then, Lossy join decomposition

$R_1 \bowtie R_2 \bowtie \dots \bowtie R_n \subset R$ (not possible)



Topic : Natural Join ($\bowtie$)

Natural Join($\bowtie$) is a derived Relational Algebra operation, which is derived using three basic Relational Algebra operation

- Projection (π)
- Selection(σ)
- Cross Product ($\times$)



Topic : Projection (π)

It is used to project the column data from a relation based on the attributes specified with projection operation.

$R(A_1, A_2, A_3, A_4, A_5, A_6)$

Resulting schema
will contain only
three attributes
 $A_1, A_2, \& A_3$

$\pi_{A_1, A_2, A_3}(R)$

Relation Name

List of attributes
required in o/p

eg:

Enroll

Sid	Cid	Branch
S ₁	C ₁	CS
S ₁	C ₂	CS
S ₂	C ₂	IT
S ₃	C ₃	CS

$$\pi_{\text{Sid, Cid, Branch}}(\text{Enroll}) \equiv (\text{Enroll})$$

o/p will be complete Enroll table!

$$\pi_{\text{Sid}}(\text{Enroll}) = \begin{array}{|c|} \hline \text{Sid} \\ \hline S_1 \\ S_1 \\ S_2 \\ S_3 \\ \hline \end{array} = \begin{array}{|c|} \hline \text{Sid} \\ \hline S_1 \\ S_2 \\ S_3 \\ \hline \end{array}$$

Relational Algebra query
will always produce distinct tuples.

If we do not use
Projection opⁿ then
all attributes of
the relation will be
present in o/p.

$$\pi_{\text{Cid, Branch}}(\text{Enroll}) =$$

Cid	Branch
C ₁	CS
C ₂	CS
C ₂	IT
C ₃	CS



Topic : Selection(σ)

It is used to select the tuples(records) from underlying relation based on the predicate condition specified with selection operation.

σ (Condition) (Relation_name)

eg:

Enroll

Sid	Cid	Branch
✓ S ₁	C ₁	CS
✓ S ₁	C ₂	CS
✗ S ₂	C ₂	IT
✓ S ₃	C ₃	CS

No Projection, as o/p will contain all the attributes

$\left(\sigma_{Cid=C_2} (Enroll) \right) =$

Sid	Cid	Branch
S ₁	C ₂	CS
S ₂	C ₂	IT

π_{Sid}

$\left(\sigma_{Cid=C_1 \vee Branch='CS'} (Enroll) \right) \equiv$

Sid	Cid	Branch
S ₁	C ₁	CS
S ₁	C ₂	CS
S ₃	C ₃	CS

Sid
S ₁
S ₃



Topic : Cross Product ($\times$)

/ Cartesian Join



Cross-product is a binary operation. Let R and S are any two relation, then cross product $R \times S$ will result in all attributes of R followed by all attribute of S with all possible combinations of tuples from R and S.

R

$\times$

A	B	C
1	2	3
1	3	5
3	6	9

m tuples

S

y

B	D
2	7
3	8
3	5

n tuples

$R \times S =$

$m \times n$
tuples

$x+y$ attributes

R.A	R.B	R.C	S.B	S.D
1	2	3	2	7
1	2	3	3	8
1	2	3	3	5
1	3	5	2	7
1	3	5	3	8
1	3	5	3	5
3	6	9	2	7
3	6	9	3	8
3	6	9	3	5



Topic : Natural Join ($\bowtie$)

- Natural join($\bowtie$) is a derived relational algebra operation which is derived using cross product, selection and projection as follows:

Let R and S are any two relations then,

$R \bowtie S$ = Step-1: Obtain " $R \times S$ "

Step-2: Select the tuples from " $R \times S$ " based on the equality condition on all common attributes of R and S.

Step-3: Project distinct attributes from the result of step-2.

R

A	B	C
1	2	3
1	3	5
3	6	9

S

B	D
2	7
3	8
3	9

$R \bowtie S = \pi_{A,B,C,D} (R \times S)$

$R.B = S.B$

↙

A	B	C	D
1	2	3	7
1	3	5	8
1	3	9	9

$R \times S =$

	R.A	R.B	R.C	S.B	S.D
✓	1	2	3	2	7
×	1	2	3	3	8
×	1	2	3	3	9
×	1	3	5	2	7
✓	1	3	5	3	8
✓	1	3	9	3	9
×	3	6	9	2	7
×	3	6	9	3	8
×	3	6	9	3	9

Note:

Let, $R(A, B, C)$ & $S(B, D)$

$$R \bowtie S = \pi_{A, B, C, D} \left(\sigma_{R.B = S.B} (R \times S) \right)$$

Let $R(A, \underline{B}, \underline{C})$ & $S(\underline{B}, \underline{C}, D, E)$

$$R \bowtie S = \pi_{A, B, C, D, E} \left(\sigma_{\substack{(R.B = S.B) \\ (R.C = S.C)}} (R \times S) \right)$$

Equality Condⁿ
on all common
attributes of R & S.

Note:

$R(A, B, C) \not\bowtie S(D, E)$

$\left. \begin{array}{l} \text{No Common attribute} \\ \text{b/w } R \text{ \& } S \end{array} \right\}$

$$R \bowtie S = \pi_{A, B, C, D, E} \left(\underbrace{(R \times S)}_{\substack{\text{No Common attributes} \\ \text{b/w } R \text{ and } S, \\ \therefore \text{NO Selection Condition}}} \right) = R \times S$$

Note: If there is no common attribute among two relations, then Natural Join will degenerate into Cross Product.

No Common attributes
b/w R and S ,
 $\therefore$ NO Selection Condition

$\hookrightarrow$ Hence all the tuples of $R \times S$ will be selected

and then

All attributes of $R \times S$ will be projected

$\therefore$ Result will be same as " $R \times S$ "

ex: let

$R =$

A	B
1	3
2	5
6	9

$S =$

B	C
2	8
4	6
6	9

What will be o/p of

$R \bowtie S =$

A	B	C

o/p of $R \bowtie S$ will be Empty Relation

Lossless Natural Join.

Note Consider the following relational table.

R

A	B	C
1	1	1
2	1	2
3	2	1

① If relation R is decomposed into two subrelations $R_1(AB)$ & $R_2(BC)$, then check whether the decomposition is lossless Join decomposition or not?

$R_1 =$

A	B
1	1
2	1
3	2

$R_2 =$

B	C
1	1
1	2
2	1

$R_1 \bowtie R_2 =$

A	B	C
1	1	1
1	1	2
2	1	2
2	1	1
3	2	1

Extra tuples {Spurious tuples}
Here,

$$R_1 \bowtie R_2 \supset R$$

$\therefore$ Lossy Join decomposition

- * Common attribute b/w R_1 & R_2 is B
 - * Values of B are neither unique in R_1 nor Unique in R_2
 - * B is neither a S.K of R_1 nor a S.K of R_2
- Hence Lossy Join decomposition

Note Consider the following relational table.

R

A	B	C
1	1	1
2	1	2
3	2	1

② If relation R is decomposed into two subrelations $R_1(AB)$ & $R_2(AC)$, then check whether the decomposition is lossless Join decomposition or not?

$R_1 =$

A	B
1	1
2	1
3	2

$R_2 =$

A	C
1	1
2	2
3	1

$R_1 \bowtie R_2 =$

A	B	C
1	1	1
2	1	2
3	2	1

- Common attribute b/w R_1 & R_2 is A.
- Values of A are unique in R_1 as well as in R_2
- i.e. A is a key in R_1 as well as in R_2
∴ Lossless Join decomposition

Here, $R_1 \bowtie R_2 = R$
∴ Lossless Join decomposition

Note Consider the following relational table.

R

A	B	C
1	1	2
2	1	2
3	2	1

③ If relation R is decomposed into two subrelations $R_1(AB)$ & $R_2(BC)$, then check whether the decomposition is lossless Join decomposition or not?

$R_1 =$

A	B
1	1
2	1
3	2

$R_2 =$

B	C
1	2
2	1

$R_1 \bowtie R_2 =$

A	B	C
1	1	2
2	1	2
3	2	1

- Common attribute is B.
- Values of B are not unique in R_1 but they are unique in R_2 .
- i.e., B is not a SK in R_1 , but B is a SK in R_2 .

And decomposition is lossless Join decomposition.

Here, $R_1 \bowtie R_2 = R$
∴ Lossless Join decomposition



Topic : NOTE

If relation R is decomposed into two sub-relations R_1 and R_2 then decomposition is lossless Join decomposition if and only if

① Attributes of $R_1 \cup$ Attributes of $R_2 = R$

and

② Attributes of $R_1 \cap$ Attributes of $R_2 \neq \emptyset$

{ i.e. there must be some common attributes }

and

③ (Attributes of $R_1 \cap$ Attributes of R_2) must be a Super key of either R_1 or R_2 or both ($R_1 \& R_2$)

i.e. Common attributes of $R_1 \& R_2$

H.W.
#Q.



Let $R(A, B, C, D, E)$ be the relational schema with following FD set
 $F = \{ AB \rightarrow C, C \rightarrow D, B \rightarrow E \}$

Which of the following decomposition is/are lossless join decomposition?

(i) $\{R_1(ABC), R_2(CD)\}$

(ii) $\{R_1(ABC), R_2(DE)\}$

(iii) $\{R_1(ABC), R_2(CDE)\}$

(ii) $\{R_1(ABCD), R_2(BE)\}$

THANK - YOU